

南京信息工程大学 实验（实习） 报告

实验（实习）名称 错误处理和文件读写 实验（实习）日期 2026/4/30 得分 指导教师 方巍

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一. 实验目的

了解和掌握错误处理和文件读写的基本原理和方法。

二. 实验内容

课本第五章 例5-2 5-4 5-7 5-11 5-14

三. 实验步骤

0. 编写通用的 KeyBoard.java. 所有实验默认使用该类

```
1 package day0501;
2
3 import java.io.*;
4
5 public class Keyboard {
6     static BufferedReader br = new BufferedReader(new
InputStreamReader(System.in));
7
8     public static int getInteger() {
9         try {
10             return (Integer.valueOf(br.readLine().trim()).intValue());
11         } catch (Exception e) {
12             e.printStackTrace();
13             return 0;
14         }
15     }
16
17     public static String getString() {
18         try {
19             return br.readLine();
20         } catch (IOException e) {
21             return "0";
22         }
23     }
24 }
25
```

1. 5-2

- 代码

```
1 package day0501;
2
3 public class ExceptionTester1 {
4     public static void main(String[] args) {
5         System.out.println("输入第一个数: ");
6     }
7 }
```

```

6         int a = Keyboard.getInteger();
7         System.out.println("输入第二个数: ");
8         int b = Keyboard.getInteger();
9         System.out.print("a / b = ");
10        int res = a / b;
11        System.out.println(res);
12    }
13 }
14

```

- 结果

```

PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0501> java .\ExceptionTester1.java
输入第一个数:
143
输入第二个数:
24
a / b = 5
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0501> java .\ExceptionTester1.java
输入第一个数:
140
输入第二个数:
abc
java.lang.NumberFormatException: For input string: "abc"
    at java.base/java.lang.NumberFormatException.forInputString(NumberFormatException.java:67)
    at java.base/java.lang.Integer.parseInt(Integer.java:527)
    at java.base/java.lang.Integer.valueOf(Integer.java:874)
    at day0501.Keyboard.getInteger(Keyboard.java:10)
    at day0501.ExceptionTester1.main(ExceptionTester1.java:8)
    at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:104)
    at java.base/java.lang.reflect.Method.invoke(Method.java:565)
    at jdk.compiler/com.sun.tools.javac.launcher.SourceLauncher.execute(SourceLauncher.java:264)
    at jdk.compiler/com.sun.tools.javac.launcher.SourceLauncher.run(SourceLauncher.java:138)
    at jdk.compiler/com.sun.tools.javac.launcher.SourceLauncher.main(SourceLauncher.java:76)
a / b = Exception in thread "main" java.lang.ArithmeticException: / by zero
    at day0501.ExceptionTester1.main(ExceptionTester1.java:10)
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0501>

```

2. 5-4

- 代码

```

1 package day0502;
2
3 public class ExceptionTester3 {
4     public static void main(String[] args) {
5         int a = 0, b = 0, res = 0;
6
7         try {
8             System.out.println("输入第一个数: ");
9             a = Integer.valueOf(Keyboard.getString().trim()).intValue();
10            System.out.println("输入第二个数: ");
11            b = Integer.valueOf(Keyboard.getString().trim()).intValue();
12            res = a / b;
13        } catch (NumberFormatException e) {
14            System.out.println("Invalid input");
15            System.exit(-1);
16        } catch (ArithmeticException e) {
17            System.out.println("Division by zero");
18            System.exit(-1);
19        }
20        System.out.println("a / b = " + res);
21    }
22 }
23

```

- 结果

```
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0502> java .\ExceptionTester3.java
输入第一个数:
143
输入第二个数:
0
Division by zero
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0502>
```

3.5-7

- 代码

```
1 // DivideByZeroException.java
2 package day0503;
3
4 public class DivideByZeroException extends ArithmeticException {
5     public DivideByZeroException() {
6         super("Division by zero");
7     }
8 }
9
```

```
1 // DivideByZeroExceptionTester.java
2 package day0503;
3
4 public class DivideByZeroExceptionTester {
5     private static int quotient(int numerator, int denominator) throws
DivideByZeroException {
6         if (denominator == 0) {
7             throw new DivideByZeroException();
8         }
9         return numerator / denominator;
10    }
11
12    public static void main(String[] args) {
13        int a = 0, b = 0, res = 0;
14        try {
15            System.out.println("输入第一个数: ");
16            a = Integer.valueOf(Keyboard.getString().trim()).intValue();
17            System.out.println("输入第二个数: ");
18            b = Integer.valueOf(Keyboard.getString().trim()).intValue();
19            res = quotient(a, b);
20        } catch (NumberFormatException e) {
21            System.out.println("Invalid input");
22            System.exit(-1);
23        } catch (DivideByZeroException e) {
24            System.out.println(e.toString());
25            System.exit(-1);
26        }
27        System.out.println("a / b = " + res);
28    }
29 }
30
```

- 结果

```
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0503> java .\DivideByZeroExceptionTester.java
输入第一个数:
20248329
输入第二个数:
0
day0503.DivideByZeroException: Division by zero
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0503>
```

3.5-11

- 代码

```
1 package day0504;
2
3 import java.io.*;
4
5 public class FileWriterTester1 {
6     public static void main(String[] args) throws IOException {
7         String filename = "./Hello.txt";
8         FileWriter fw = new FileWriter(filename);
9         fw.write("Hello world!\n");
10        fw.write("--From 202483290476 张正珂\n");
11        fw.close();
12    }
13 }
14
```

- 结果

```
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0504> cat ./Hello.txt
现在是空的
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0504> java ./FileWriterTester1.java
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0504> cat ./Hello.txt
Hello World!
--From 202483290476 张正珂
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0504>
```

3.5-14

- 代码

```
1 package day0505;
2
3 import java.io.*;
4
5 public class BufferedReaderTester {
6     public static void main(String[] args) throws IOException {
7         String filename = "./Hello.txt";
8         String line;
9
10        try {
11            BufferedReader br = new BufferedReader(new
12            FileReader(filename));
13            line = br.readLine();
14            while (line != null) {
15                System.out.println(line);
16                line = br.readLine();
17            }
18            br.close();
19        } catch (IOException e) {
20            System.out.println("张正珂没找到文件" + filename + "在哪儿!");
21        }
22    }
23 }
```

- 结果

```
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0505> cat .\Hello.txt
Hello World!
--From 202483290476 张正珂
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0505> java .\BufferedReaderTester.java
Hello World!
--From 202483290476 张正珂
PS E:\Coding Projects\Programme-Learning-Repo\Java\Day05\day0505> █
```

四. 体会和总结

通过本次“错误处理和文件读写”实验，我对 Java 程序健壮性的实现方式有了更直观的认识。总体来看，本次实验帮助我把课堂中的概念转化为可运行代码，提升了调试与排错能力。后续我会继续规范变量命名与异常提示信息，并尝试使用 `try-with-resources` 优化资源管理，使程序更简洁、更安全。